

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

eveth fine it's ok no problems
i haven't started
it's okay no fine great
no concerns nope reading materials
a little bit tired

Today's agenda

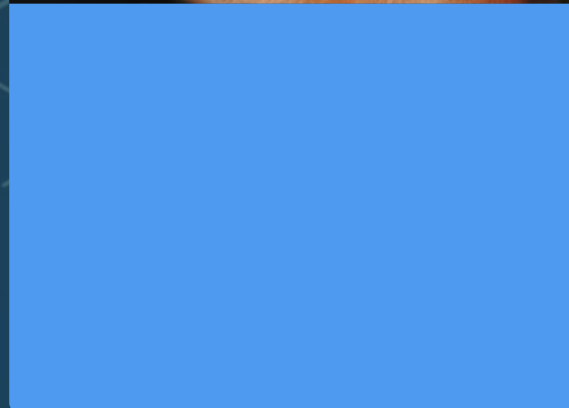
- Recall
- Architectural Patterns
- MVC, MVP, MVVM
- Comparison
- Layered architecture pattern
- Live coding: MVC with servlet
- QUIZ

What do you remember from the last session?

we're gona have 1-1 mids
architectural patterns
schemes
layers
sdkman
gg
chill
java webapps
mvc
mvvm
deadlines almost kill
course structure
course proj explanation
plans for the course
mvp

Java | Python | C | Not Found ???

17 ✓



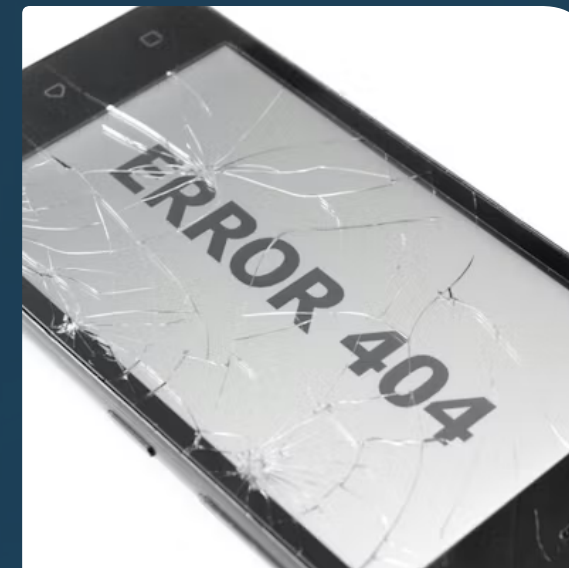
8 ✓



9 ✓



5 ✓



Benefits of Java language in Web Development ?

Scalability

I don't remeber

security

large community

Platform independence

platform independ

Secure

some pretty features

Benefits of Java language in Web Development ?

Enterprise-grade
security

rich ecosystem

secure, scalable, fast

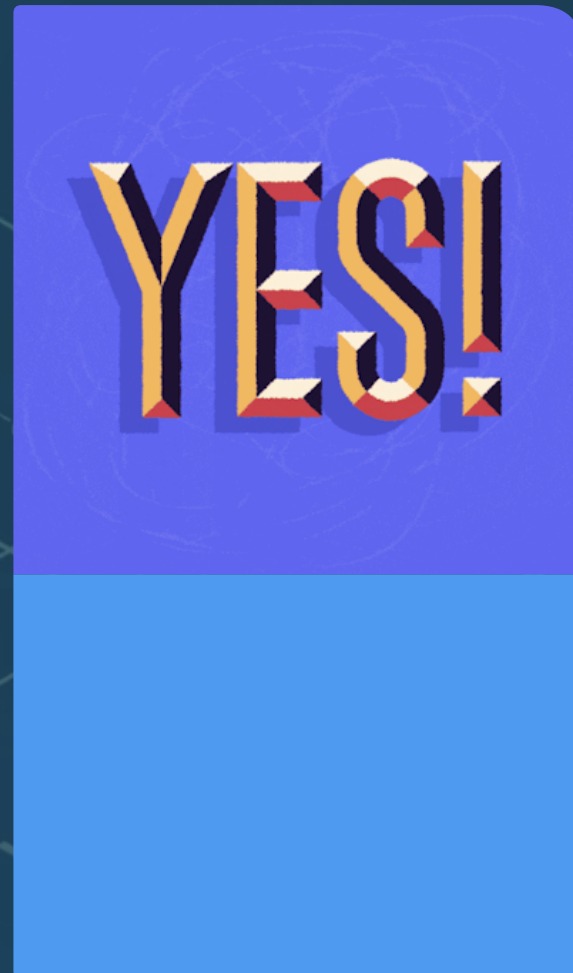
Benefits

- Community (Experience)
- Libs \ frameworks (Opensource)
- Huge legacy supported by giants (Demand)

Architecture Advanced

Have you started the theory of architecture advanced?

7



3



6



????????????????

Live coding

MVC with servlet

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026/tree/main/ws-02-architecture-advanced>

What is the difference between Architectural pattern and Design patterns?

architecture - code;
design - ui

architecttural about
backend, desing about
frontend

Scale

scope and level of
abstraction

Architectural choices are
fundamental and hard to
change once implemented

Architectural pattern -
Microservices Event-
Driven, Hexagonal

architectural pattern define the
overall structure of an app,
while design solves specific
design issues within that
structure i guess

Architectural patterns
structure an entire
application, while design
patterns solve specific
problems within that
structure

What is the difference between Architectural pattern and Design patterns?

High-level system structure versus code-level solutions

Architectural patterns define the overall system structure, while design patterns solve specific problems within it

design patterns dictate how individual classes and objects collaborate

Defines the overall system's structure vs solves a specific design problem within the system

Architectural pattern give us how we work with datas

architectural pattern is the layout plan for your entire application, while a design pattern is a for solving a specific problem

arch - global stucture + high-level subsys organization; design - localized problems within subsys

Architecture

- address a myriad of challenges in software engineering, such as computer hardware performance restrictions, high availability, and business risks
- deals with the overall structure of the application

Design Pattern

- recommended solutions
- best practice

MVC, MVP, MVVM

Decipher abbreviations MVC, MVP, MVVM

model view controller

model view
controller/presenter/view-model

MODEL VIEW VIEW
MODEL

Model View Controller,
Model View Presenter,
Model View View-Model

Model-View-Controller
Model-View-Presenter
Model-View-ViewModel

model view controller;
model view presenter;
model view viewmodel

model view controller
model view presenter
model view view model

Model view controller
Model view presenter
Model view viewmodel

Decipher abbreviations MVC, MVP, MVVM

Model View
Controller/Presenter/View
wModel

MVC Model-View-Controller
MVP Model-View-Presenter
MVVM Model-View-
ViewModel

MVC

MVC (Model-View-Controller)

- **Primary Use:** Server-side web applications and traditional desktop software.
- **Where it's used:** Classic web frameworks like Ruby on Rails, ASP.NET MVC, Laravel (PHP), and Django. It's also used in iOS development via UIKit.
- **How it works:** The Controller receives a web request from a browser, fetches data from the Model, and renders an HTML View back to the user.

MVP

MVP (Model-View-Presenter)

- **Primary Use:** Legacy Android applications, Windows Forms (WinForms), and older desktop GUIs.
- **Where it's used:** Popular in Android apps built prior to 2017–2018 before modern architecture components like Jetpack ViewModel were introduced.
- **How it works:** The Presenter acts as a middleman. When a user clicks a button on the View, the View notifies the Presenter, which updates the Model and explicitly instructs the View to update its UI components.

MVVM

MVVM (Model-View-ViewModel)

- **Primary Use:** Modern mobile app development (Android and iOS) and reactive desktop/frontend frameworks.
- **Where it's used:** Modern Android apps (recommended by Google using Jetpack), iOS apps using SwiftUI, WPF/.NET applications, and modern web frameworks like Vue.js or Knockout.js.
- **How it works:** Uses **data binding**. Instead of the ViewModel telling the View to repaint itself, the View automatically observes streams of state in the ViewModel and updates itself whenever data changes.

Place from old to new and from min benefits to max



MVC

Benefits

- Division into modules organizing the application code,
- Separation of business logic from view,
- The model is not dependent on the view,
- Makes it easier to find a specific part of the code,
- Easier expansion through modular construction,
- Prevents clutter in the code,
- Facilitates teamwork.

MVC

Disadvantages

- Complicated division into modules increasing the complexity of the system,
- Due to the specificity of the Android system architecture (life and activity cycle), the View and Controller layers are often implemented in an activity, a fragment, so there is no clear separation between them (therefore only the model layer is able to perform unit testing),
- Dependence of the view on the model,
- Difficult to test views,
- The pattern can be used in small projects with a small number of screens, where, if necessary, the business logic is tested.

MVP

Benefits

- Division into modules organizing the application code,
- Separation of business logic from view,
- The model is not dependent on the view,
- Makes it easier to find a specific part of the code,
- Easier expansion through modular construction,
- Facilitates teamwork,
- Is used to solve the problem of large activities, fragments (which have too much responsibility and a lot of code).

MVP

Disadvantages

- Due to the lifecycle of a fragment or activity, or other user interface components, a reference memory leak may appear for a view that does not exist anymore in the presenter. In such a case, care should be taken to adequately support lifecycle methods,
- It often happens that the implementation of a pattern requires the creation of additional classes and interfaces (often with similar content), which is often associated with redundant code,
- Too complicated for simple, small applications,
- Requires familiarizing employees with the pattern, which results in higher costs,
- Presenter has a reference to the view so if we want to use it elsewhere we need to implement that view in for example fragment. This can result in many unnecessary empty functions implemented,
- Tight coupling between

MVVM

Benefits

- Solves the memory leak problem that appears in the MVP pattern (the model view does not have a reference to the view layer),
- No duplication of data,
- Independence of logic from the way data is displayed,
- Thanks to the separation of tasks in different layers, the pattern can be used to increase the readability of the code and the possibility of independent testing without the use of additional dependencies,
- No interfaces defining views like in MVP,
- The data binding mechanism increases the flexibility of the view on modifications, which makes the MVVM pattern work well in large dynamically changing projects.

MVVM

Disadvantages

- Each view screen has a dedicated ViewModel, which translates into a greater number of classes (compared to the MVP, there are still fewer of them),
- The view layer must ensure proper binding of variables and methods for each required view element, as well as observing the state of the ViewModel,
- Creating a correct and complete ViewModel requires analyzing the view layer in terms of the required data and possible states,
- Proper handling of LiveData (live events from viewModel to view layer) requires more boilerplate code,
- In some cases data binding mechanism can get complex. You need to be careful when you modify your UI layout files,
- When used with data binding you need to be aware of clearing binding every time user leaves the screen, otherwise memory leaks might occur.

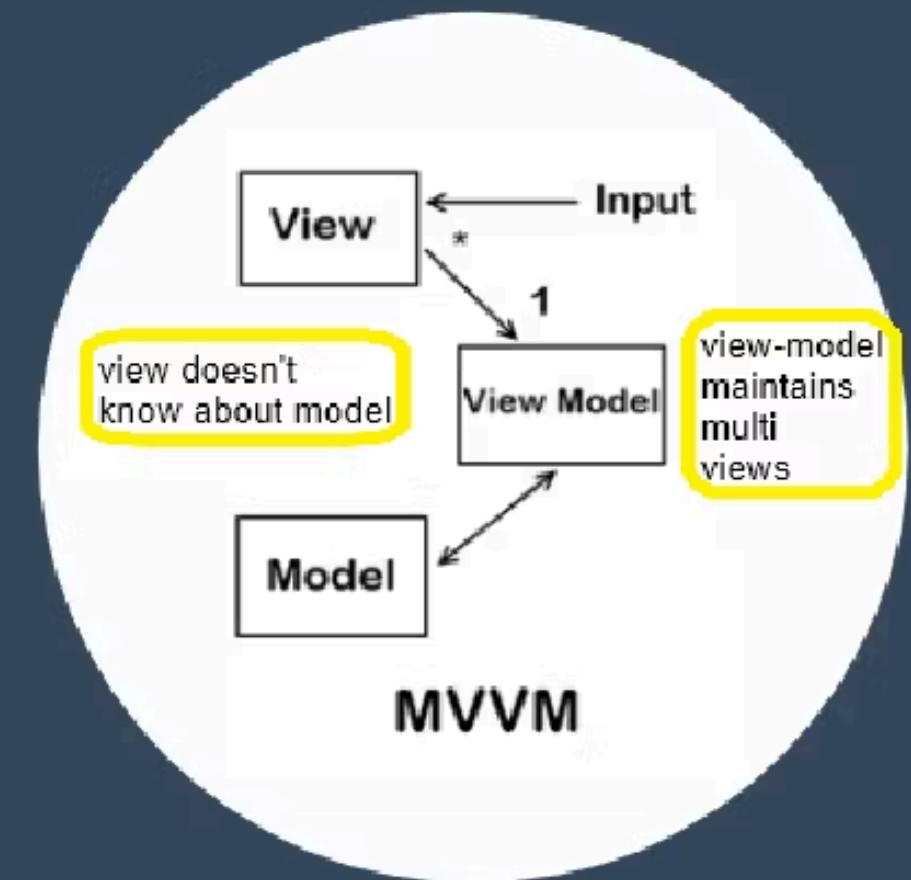
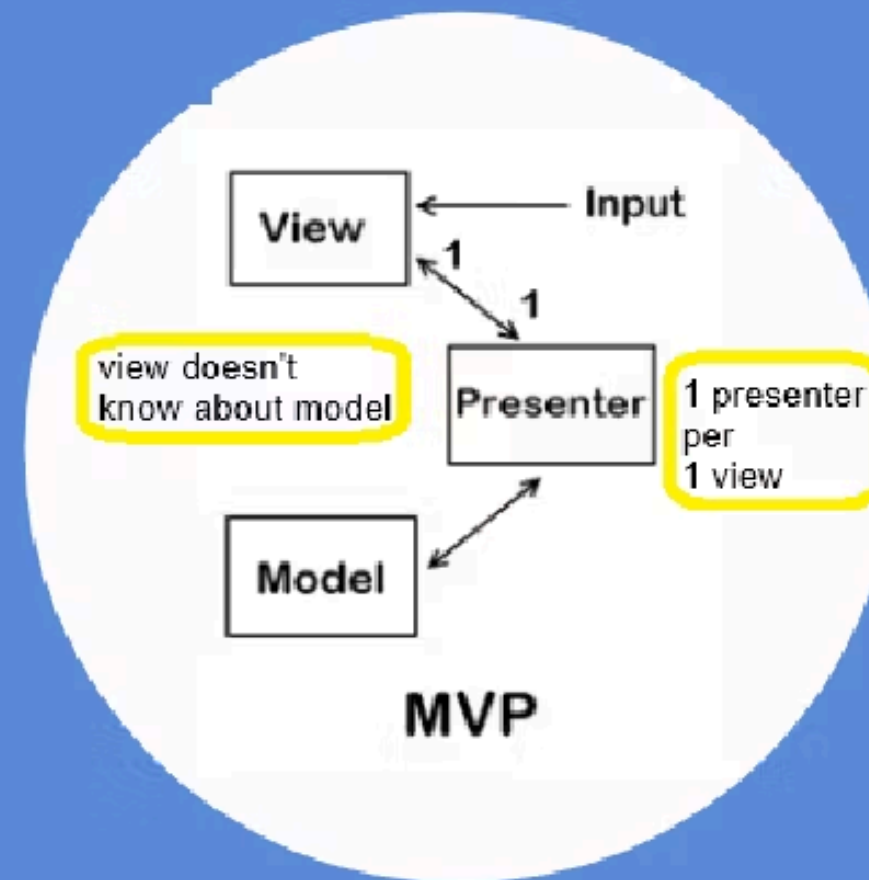
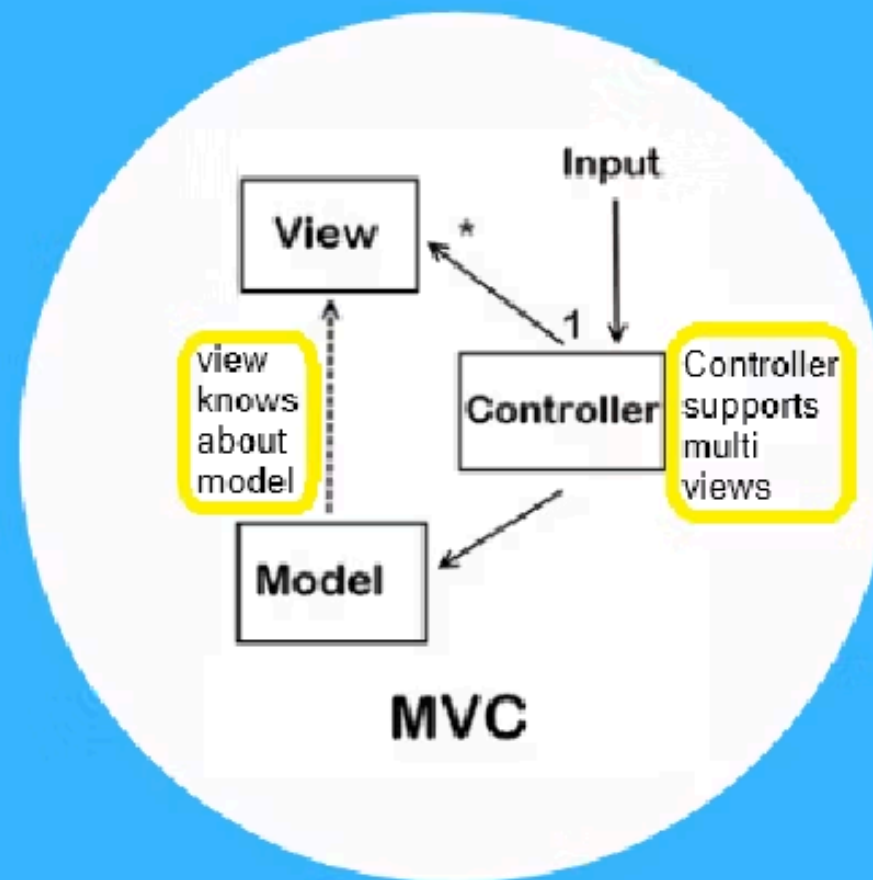
Summary

In MVC the Controller handles events, manipulates the Model, which contains business logic, and the View displays data from the Model. One Controller supports several Views.

In MVP, the data from the Model is passed to the Presenter and not directly to the View, and the Presenter passes it to the View. One presenter can refer to one View.

In MVVM, the data from the Model is passed to the ViewModel, which can support multiple Views. The View knows nothing about the ViewModel, it only requires the data you need.

MVC VS MVP VS MVVM



Additional explanation

<https://itcraftapps.com/blog/mvc-mvp-and-mvvm-architecture-pattern-introduction/#:~:text=In%20MVP%2C%20the%20data%20from,which%20can%20support%20multiple%20Views.>

Layered architecture pattern

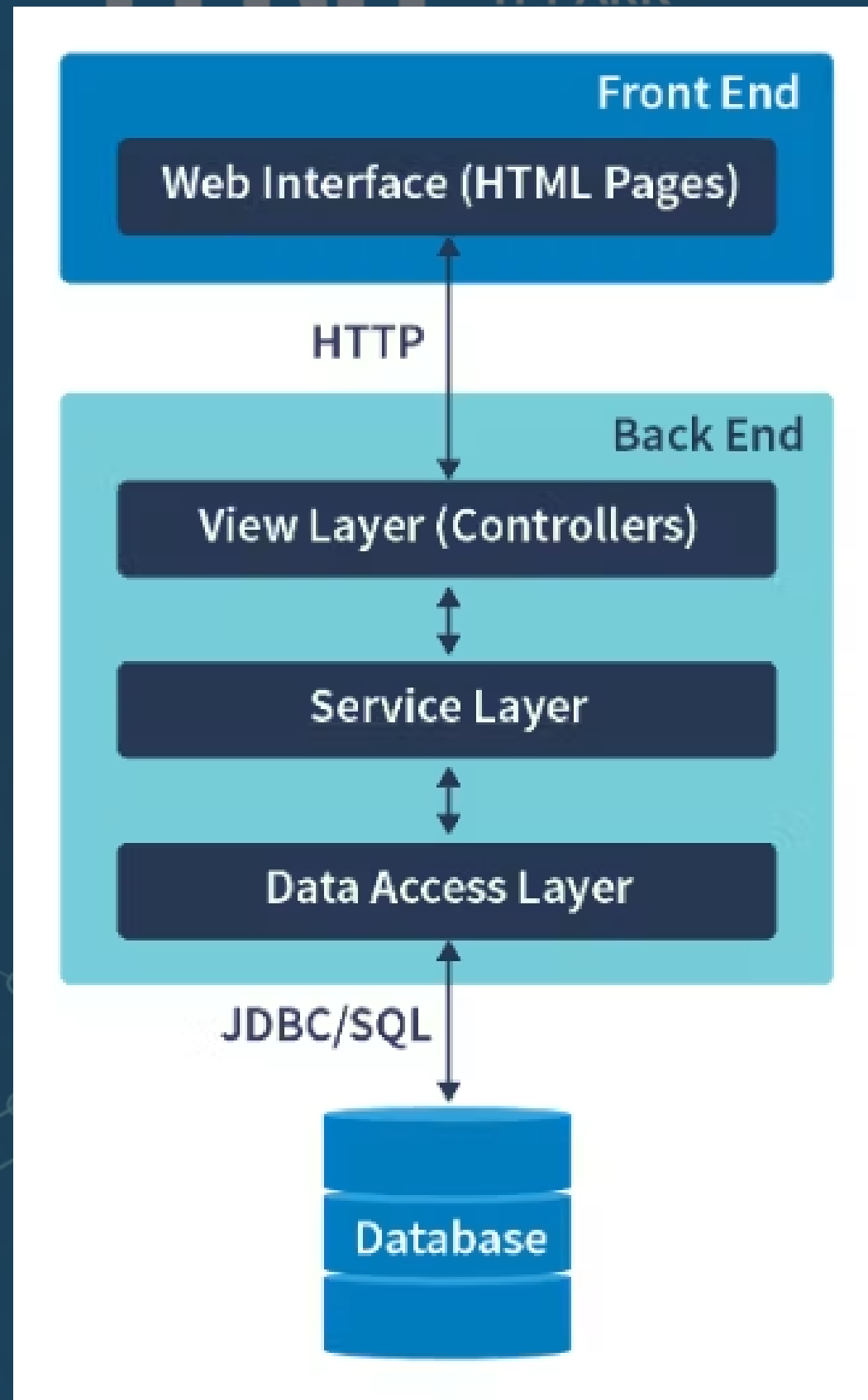
What are the layers in Layer Design ???

thing that are being
designed

layer 1 layer 2 layer 3

Presentation, Business
Logic, Data Access,
Database

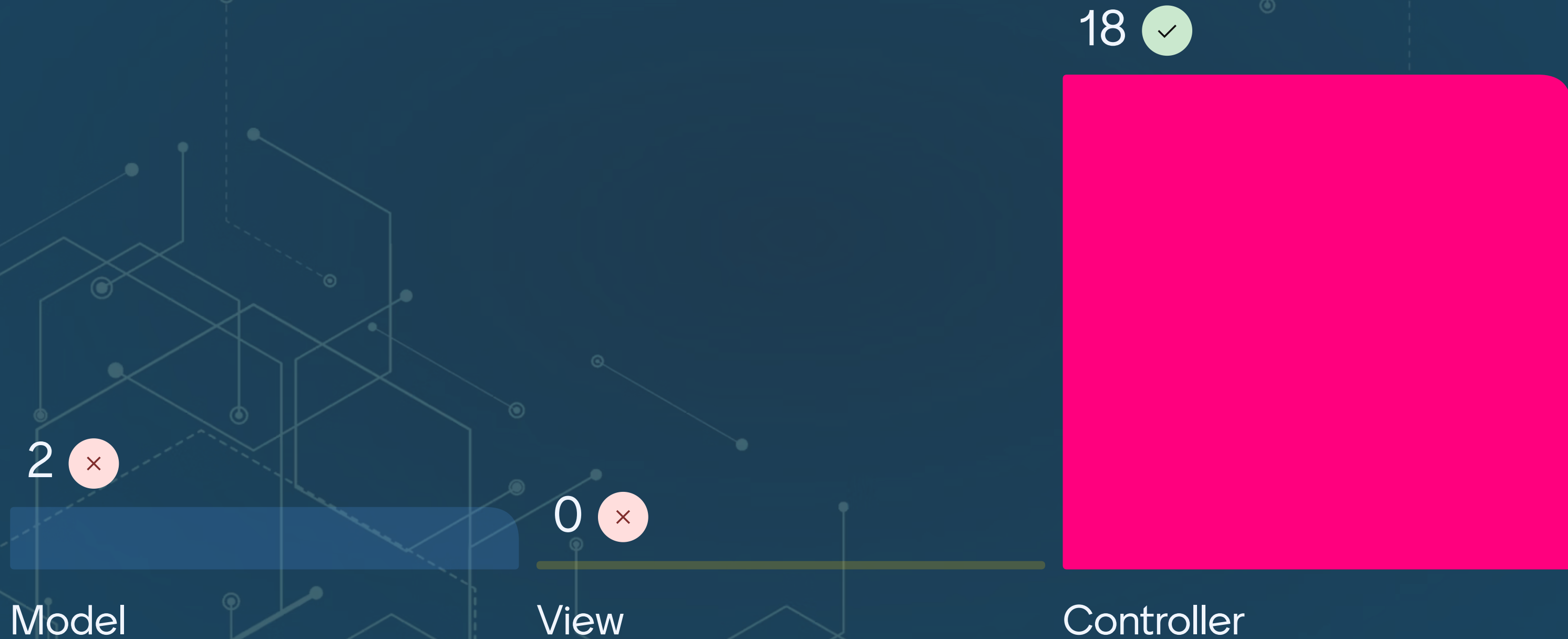
Presentation Layer



QUIZ

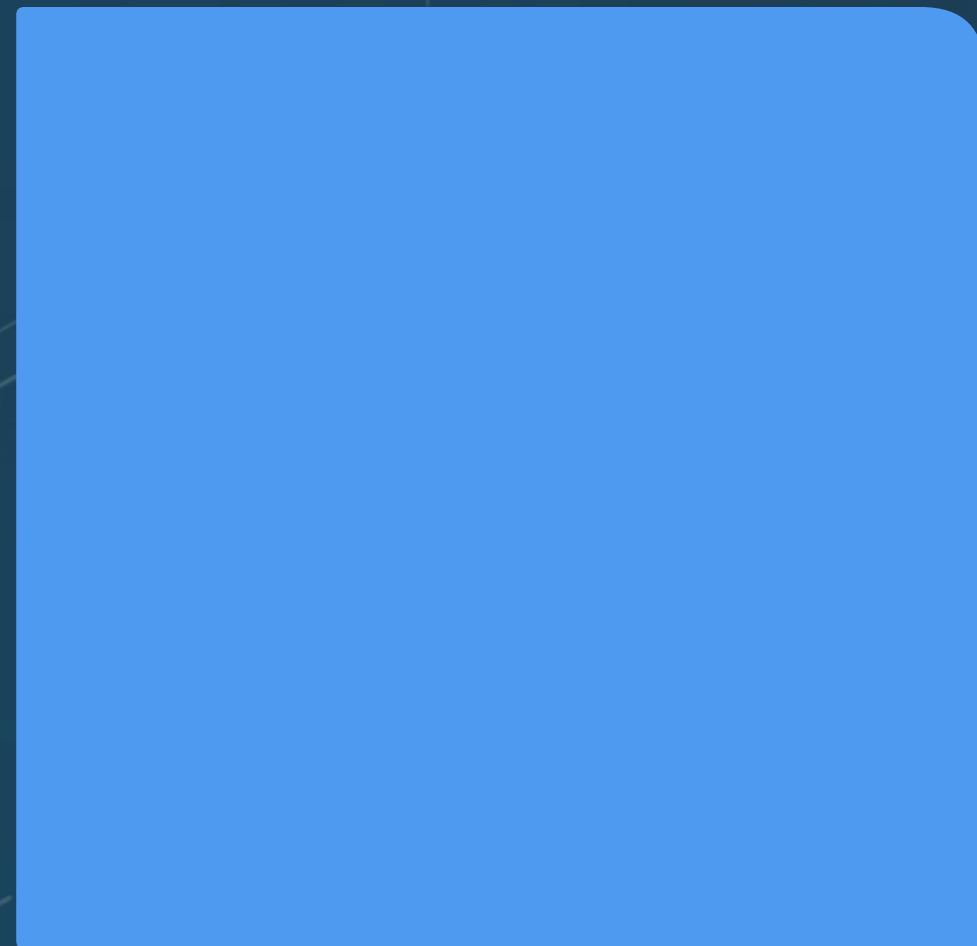


Who is receiving requests in MVC?



Who is keeping business logic in MVC?

18 ✓



Model

1 ✗



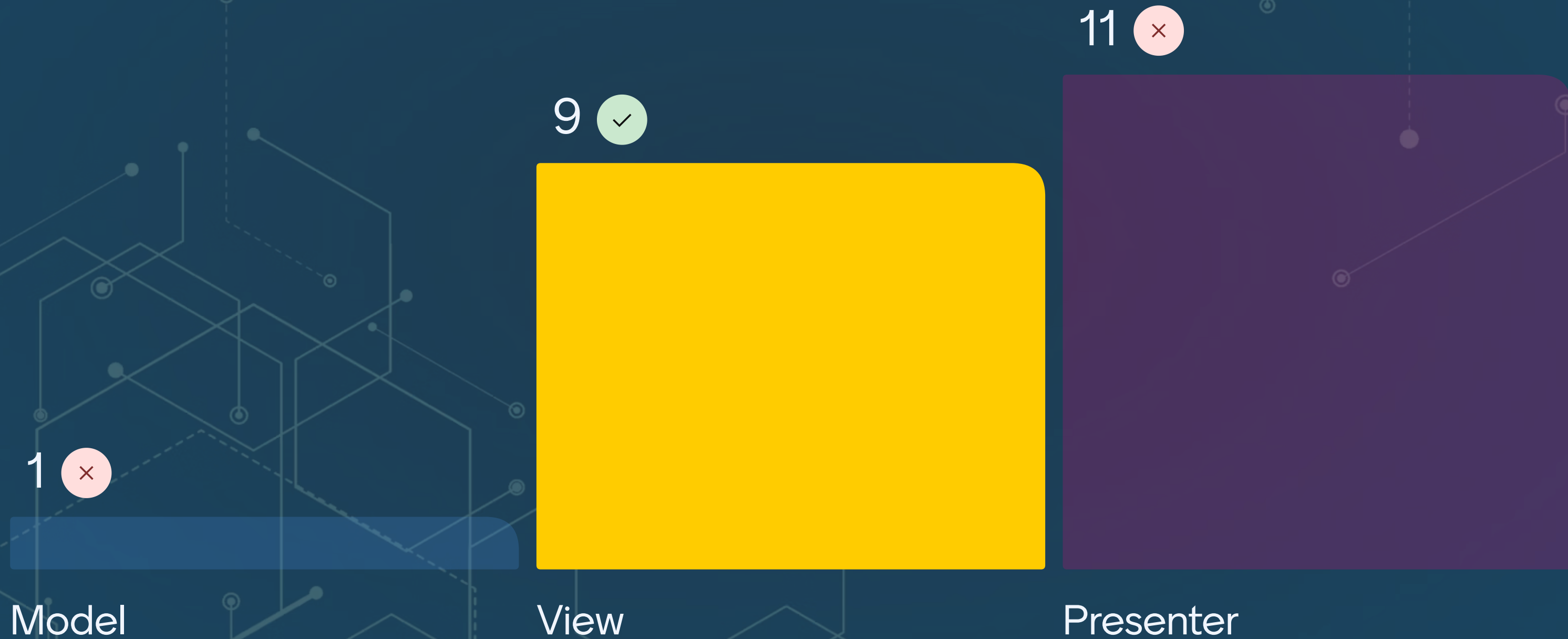
View

2 ✗



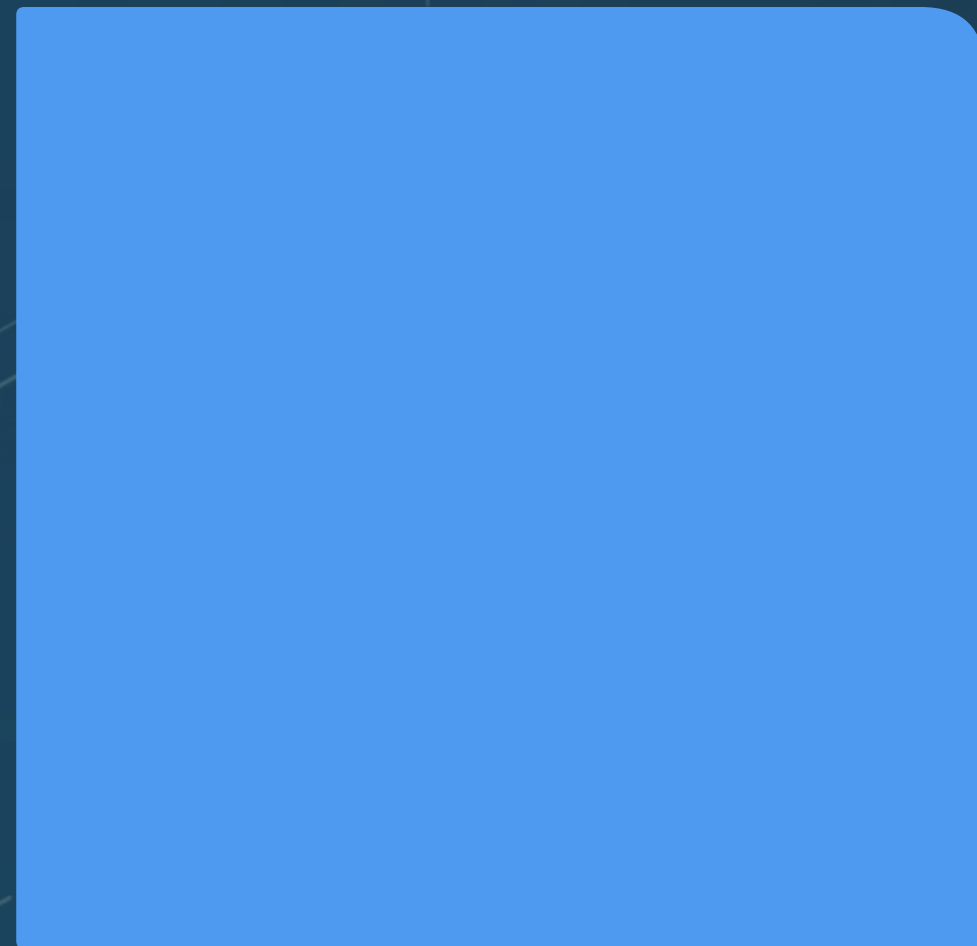
Controller

Who is receiving requests in MVP?



Who is keeping business logic in MVP?

14 ✓



Model

0 ✗

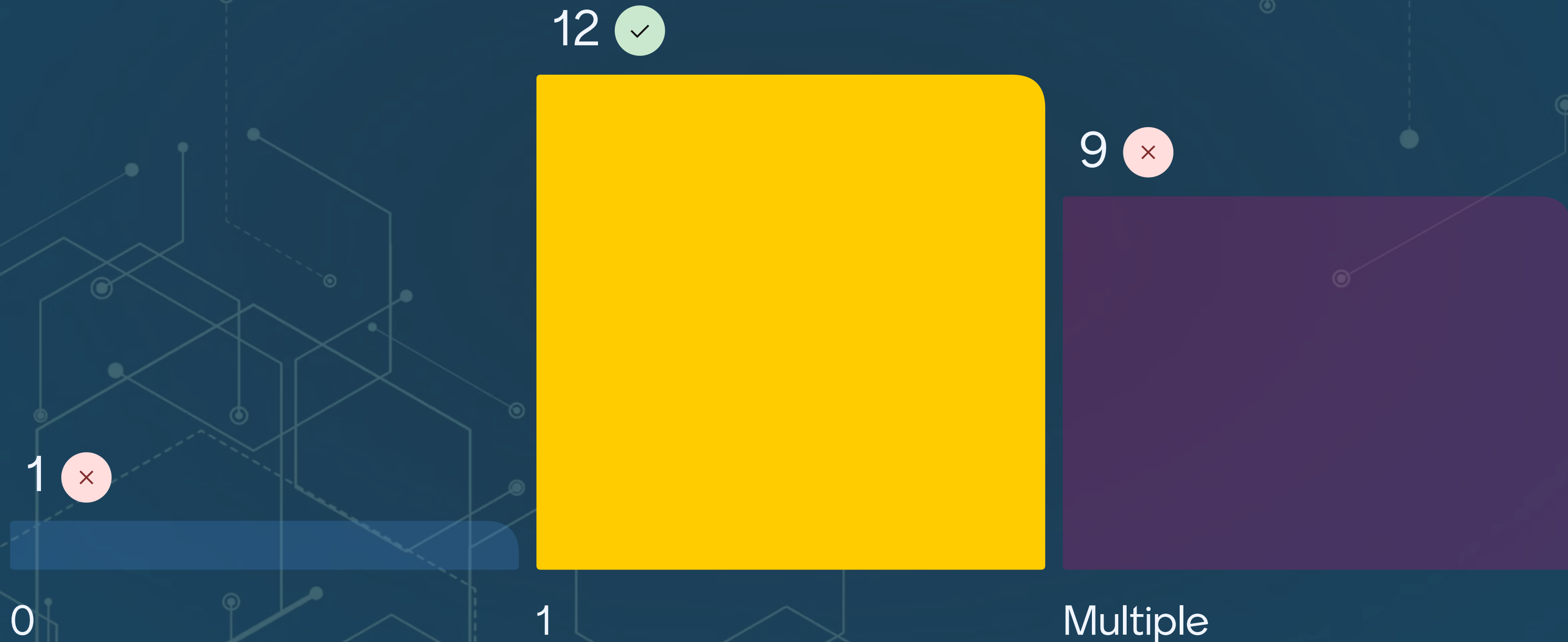
View

8 ✗



Presenter

How many views can be referred to Presenter in MVP



How many views can be referred to ViewModel in MVVM

1 ✗

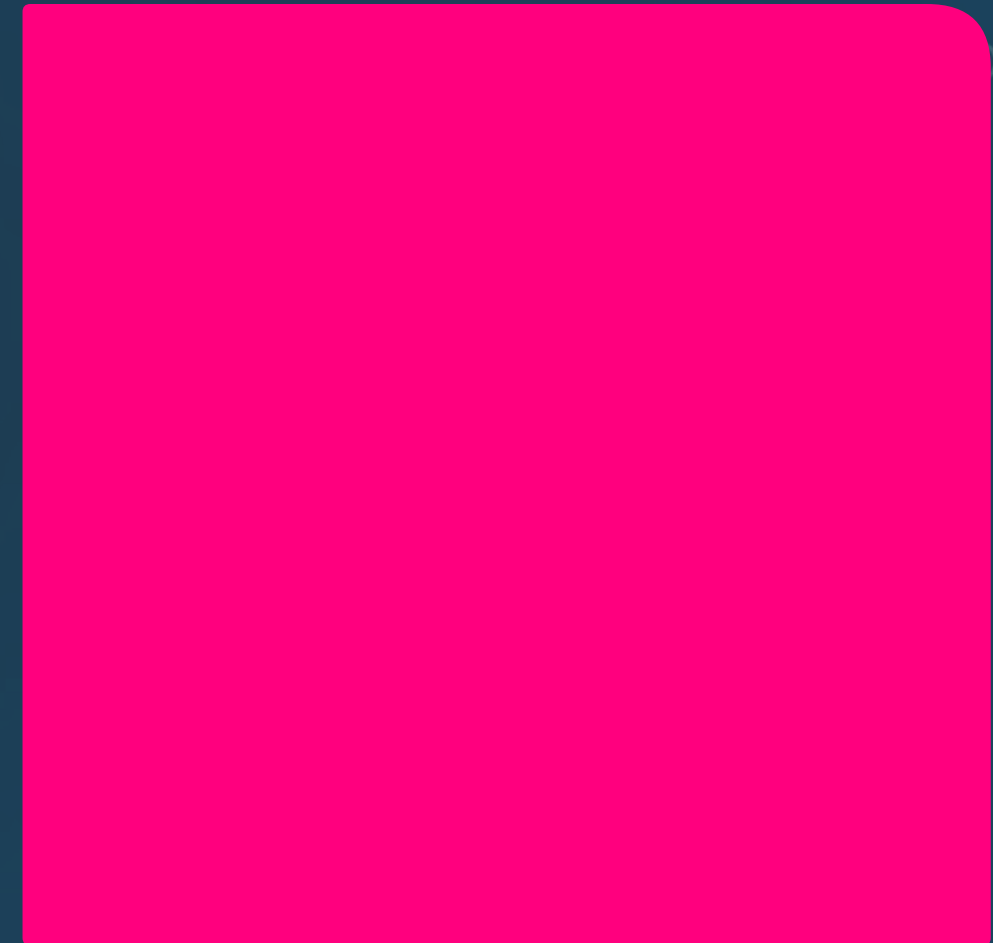
0

2 ✗

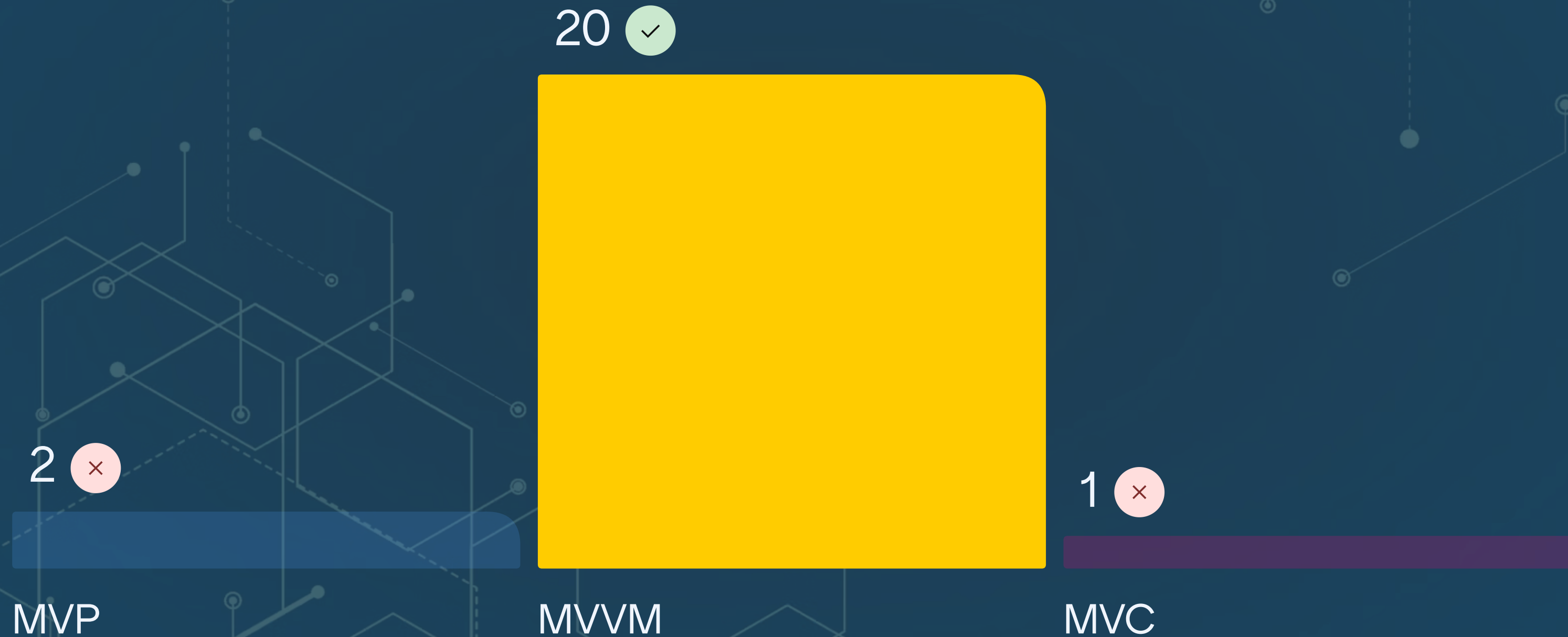
1

20 ✓

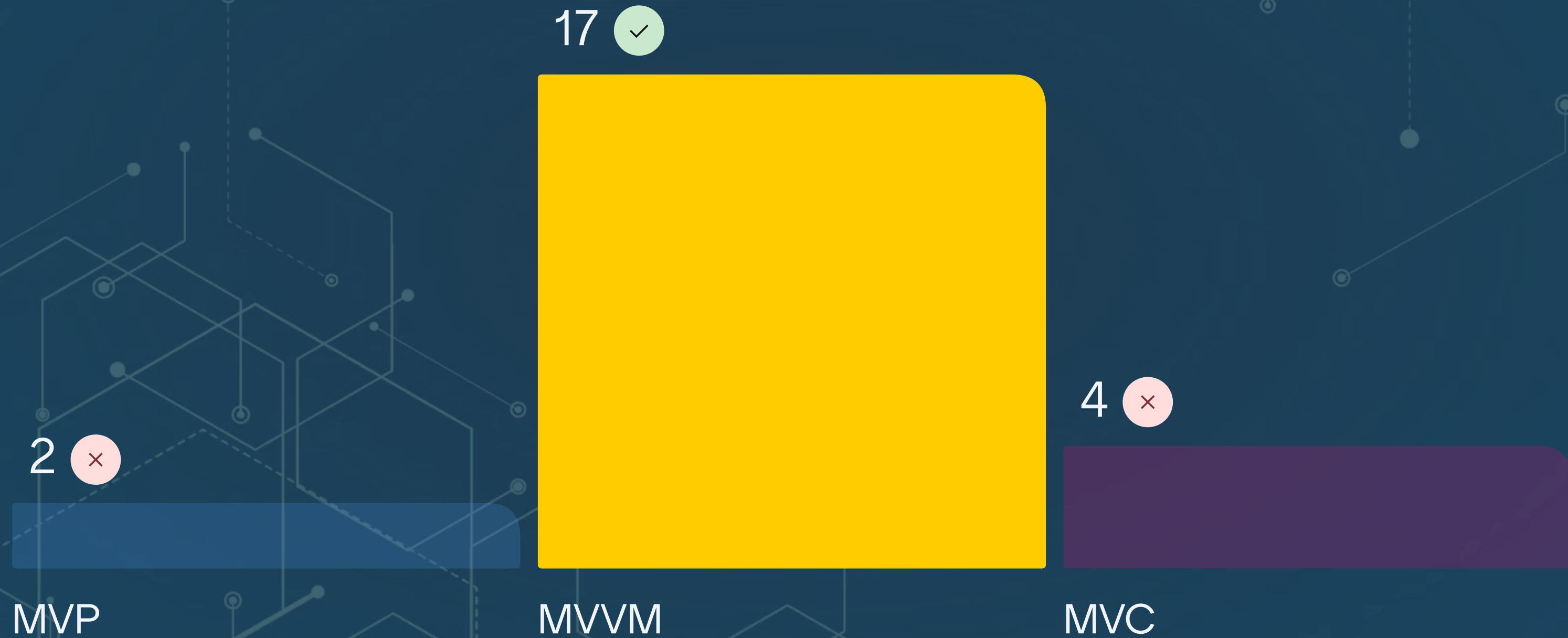
Multiple



Choose most recent architectural pattern



Choose most beneficial architectural pattern



In which architectural pattern View knows about Model

1 ✗

MVP

0 ✗

MVVM

22 ✓

MVC



How many structural layers in 'Layered Design' ?

22 ✓

0

✗

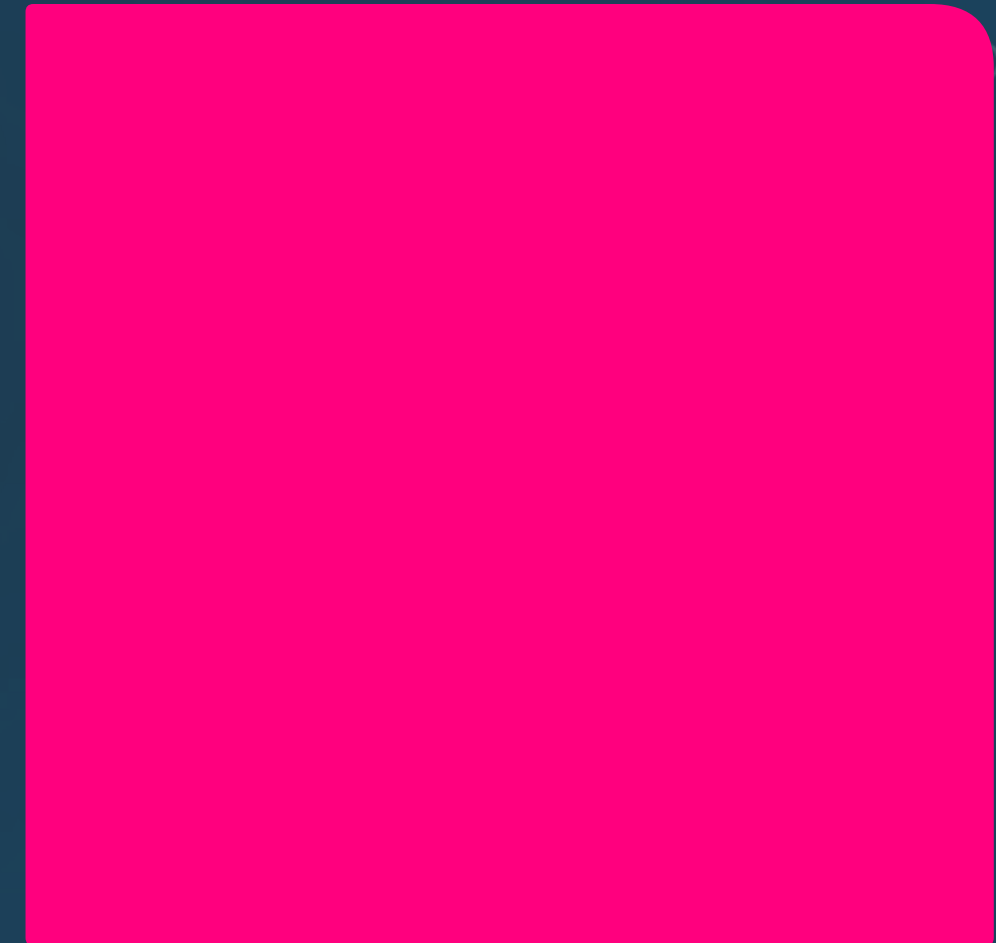
1

0

✗

2

3



Quiz leaderboard



Questions?



Next time: Database Layer Design

